

CHAPTER 7: CONFOUNDING

Causal Diagrams, Exchangeability, and Adjustment Methods

Applied Use Case: *Did the Retention Offer Actually Reduce Churn?*

Executive Summary & Chapter Overview

Chapter 7 of *Causal Inference: What If* (Hernán & Robins) provides the formal structural and counterfactual definition of **Confounding**—the primary threat to causal inference in observational data. Confounding arises when treatment A and outcome Y share **common causes** L (or unmeasured U), opening a non-causal **backdoor path** $A \leftarrow L \rightarrow Y$ that induces association without causation (Section 7.1, pp. 85–87).

In our retention use case, if high-risk customers ($L = 1$, e.g., frequent complaints) are more likely to receive a retention offer ($A = 1$) AND intrinsically more likely to churn ($Y = 1$), the raw observational association conflates the true effect of the offer with the baseline risk of the customers.

This document synthesizes Chapter 7’s core theoretical pillars:

1. **Structural Definition of Confounding:** Presence of open backdoor paths due to common causes (Section 7.1–7.2, pp. 85–89).
2. **The Backdoor Criterion:** Graphical rules for identifying sufficient adjustment sets L (Section 7.3, pp. 89–92).
3. **Confounder vs. Non-Confounder:** Why traditional statistical definitions fail and why M -bias occurs (Section 7.4, pp. 92–95).
4. **Single-World Intervention Graphs (SWIGs):** Explicit visual representation of counterfactual outcomes Y^a (Section 7.5, pp. 95–96).
5. **Confounding Adjustment Methods:** G-methods (IP Weighting, Standardization, G-estimation) vs. Conventional Methods, and why post-treatment variables cannot be adjusted for (Section 7.6, pp. 96–98).
6. **Front-Door Criterion & Negative Controls:** Advanced identification strategies when common causes U are unmeasured (Technical Points 7.3 & 7.4, pp. 99–101).

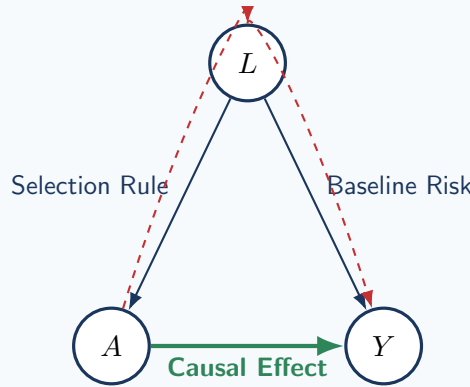
1. The Structure of Confounding & Backdoor Paths

Definition (Section 7.1, p. 85): Confounding is the bias resulting from the existence of shared common causes of treatment A and outcome Y .

In a causal DAG, any path between A and Y that starts with an arrow pointing INTO treatment ($A \leftarrow \dots \rightarrow Y$) is called a **backdoor path**. An open backdoor path allows spurious association to flow between A and Y , breaking marginal exchangeability:

$$\Pr[Y^{a=0} = 1 \mid A = 1] \neq \Pr[Y^{a=0} = 1 \mid A = 0]$$

Figure 7.1: Classical Confounding Structure in CRM Data



- **A (Retention Offer):** 1 = Discount sent, 0 = No offer.
- **Y (Customer Churn):** 1 = Canceled subscription, 0 = Retained.
- **L (Baseline Complaint History / Risk Score):** Common cause driving both offer assignment ($L \rightarrow A$) and baseline risk of churn ($L \rightarrow Y$).
- **Red Dashed Path ($A \leftarrow L \rightarrow Y$):** Open backdoor path creating non-causal association.

2. Confounding and Exchangeability

Section 7.2 (pp. 87–89): In an observational study, marginal exchangeability $Y^a \perp\!\!\!\perp A$ rarely holds due to confounding. However, if all common causes L are measured, we can achieve **Conditional Exchangeability**:

$$Y^a \perp\!\!\!\perp A \mid L \quad \text{for all } a$$

Conditional exchangeability means that within any stratum of $L = l$ (e.g., among customers with exactly 2 complaints in the last quarter), those who received the offer ($A = 1$) and those who did not ($A = 0$) have identical counterfactual risk distributions:

$$\Pr[Y^{a=1} = 1 \mid A = 1, L = l] = \Pr[Y^{a=1} = 1 \mid A = 0, L = l] = \Pr[Y^{a=1} = 1 \mid L = l]$$

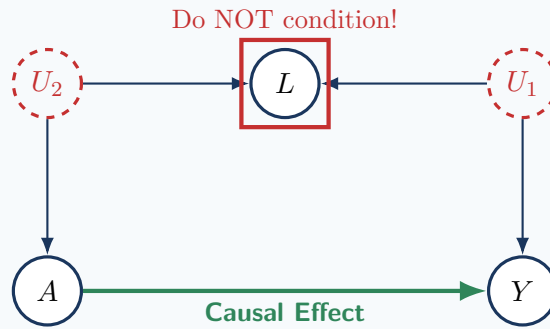
When conditional exchangeability holds, L is a **sufficient set for confounding adjustment**.

3. The Backdoor Criterion & M-Bias

Section 7.3 (pp. 89–92): To determine whether a set of variables L is sufficient to eliminate confounding, Pearl’s **Backdoor Criterion** requires that L :

1. Blocks all backdoor paths between A and Y .
2. Contains no descendants of treatment A .

Figure 7.4: M-Bias Structure – Conditioning on a Non-Confounder Induces Bias



- **Unconditioned Graph:** The backdoor path $A \leftarrow U_2 \rightarrow L \leftarrow U_1 \rightarrow Y$ is **BLOCKED** at baseline because L is a **collider** ($U_2 \rightarrow L \leftarrow U_1$). Marginal exchangeability holds ($Y^a \perp\!\!\!\perp A$).
- **M-Bias Hazard:** Conditioning on L (boxed) **OPENS** the backdoor path, creating selection bias!
- **CRM Interpretation:** U_2 = Tech savviness, U_1 = Customer price sensitivity, L = App registration. L is not a cause of A or Y . Adjusting for L introduces false bias.

4. Confounding vs. Confounders

Section 7.4 (pp. 92–95): Traditional epidemiology/statistics defined a "confounder" via 3 statistical rules: (1) associated with A , (2) associated with Y conditional on A , (3) not on the causal pathway.

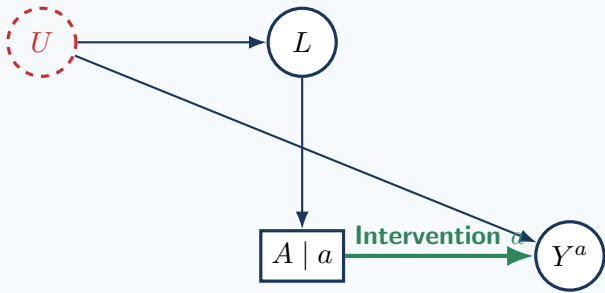
Hernán & Robins demonstrate that **statistical rules fail**:

- In Figure 7.4 (M -bias), L satisfies all 3 traditional rules, yet adjusting for L **creates bias**.
- **Structural Definition:** Confounding is a property of the causal network (presence of open backdoor paths). A variable L is a *confounder* if and only if it belongs to a sufficient set required to block open backdoor paths.
- **Surrogate Confounders (Fine Point 7.3, p. 94):** If true confounder U (e.g., unmeasured customer dissatisfaction) cannot be measured, adjusting for a proxy L (e.g., recorded support tickets) partially blocks the backdoor path $A \leftarrow U \rightarrow Y$.

5. Single-World Intervention Graphs (SWIGs)

Section 7.5 (pp. 95–96): Proposed by Richardson and Robins (2013), SWIGs unify DAGs with potential outcomes by explicitly representing counterfactual variables under a fixed intervention a .

Figure 7.9: Single-World Intervention Graph (SWIG) for Confounding



- **Node Splitting:** Treatment node is split into A (observed natural value) and a (fixed intervention value).
- **Counterfactual Outcome Y^a :** Replaces observed Y .
- **d -Separation Check:** L d -separates A from Y^a on the SWIG, visually proving $Y^a \perp\!\!\!\perp A \mid L$.

6. Methods for Confounding Adjustment

Section 7.6 (pp. 96–98): When a sufficient set L is measured, two primary analytical families eliminate confounding:

Method Category	Specific Analytical Tools	Key Characteristics & Use Case
G-Methods	• Standardization (g-formula) • Inverse Probability (IP) Weighting • G-Estimation	Estimates the marginal average causal effect $E[Y^{a=1} - Y^{a=0}]$ across the entire population. Essential for time-varying confounders.
Conventional Methods	• Stratification / Restriction • Matching • Multivariable Outcome Reg.	Estimates conditional effects within strata of L . Fails when post-treatment confounders are present (Part III).

Warning: Post-Treatment Variables Cannot Be Adjusted For (Fine Point 7.4, p. 91)

If L is a descendant of treatment A ($A \rightarrow L \rightarrow Y$), conditioning on L blocks the true causal effect of A on Y mediated through L , and may open collider paths. Confounders must always be non-descendants of treatment A .

7. Advanced Identification: Front-Door Criterion & Negative Controls

Advanced Identification Methods (Technical Points 7.3 & 7.4)

1. **Front-Door Criterion (Technical Point 7.4, p. 101):** If unmeasured U confounds $A \rightarrow Y$, but there exists a mediator M ($A \rightarrow M \rightarrow Y$) such that U does not affect M and A has no direct effect on Y except through M , the causal effect is identified via the Front-Door Formula:

$$\Pr[Y^a = 1] = \sum_m \Pr[M = m \mid A = a] \sum_{a'} \Pr[Y = 1 \mid M = m, A = a'] \Pr[A = a']$$

Retention Example: A = Offer, M = Portal Feature Usage, Y = Churn, U = Unmeasured Price Sensitivity.

2. **Negative Outcome Controls / Diff-in-Diff (Technical Point 7.3, p. 99):** Uses a control outcome N known not to be affected by A , but sharing the same unmeasured confounders U , to detect and subtract residual confounding bias.